

Project Proposal

Asynchronous Deterministic Coupling for Distributed Spatial Simulations

Hritvik

1. Problem and Motivation

Large-scale Parallel Discrete Event Simulations (PDES), particularly high-resolution discrete spatial grids (sub 1m scale), face a fundamental scalability limit: the synchronization bottleneck at partition boundaries. When simulating emergent physical interactions—such as thermal-material phase changes occurring across a spatial boundary—nodes must coordinate state transitions to ensure physical consistency.

Standard distributed architectures rely on conservative synchronization, where nodes block execution until all incoming messages from neighbors are verified. In a high-frequency environment, this assumes a "zero-latency" network which leads to significant compute-resource idling.

A practical application of this work is distributed robot fleet coordination. For example, autonomous warehouse robots or drone swarms often simulate future trajectories using spatial models. When robots operate in distributed compute environments, trajectory updates near partition boundaries require frequent synchronization. Traditional conservative PDES approaches introduce latency, limiting real-time responsiveness. A deterministic asynchronous coupling mechanism would allow robot nodes to independently compute trajectory updates while maintaining consistent shared spatial decisions, improving scalability for large autonomous fleets. [4, 5]

Research Question: Can a rule-based deterministic hierarchy enable optimistic synchronization in distributed spatial simulations without the memory-intensive overhead of state-saving snapshots?

Addressing this is critical for modern heterogeneous clusters. By removing the need for global synchronization windows, we can execute dense spatial models on distributed nodes that are limited by local memory capacity.

2. State-of-the-Art Solutions

Parallel Discrete Event Simulation systems typically rely on either conservative or optimistic synchronization mechanisms. Conservative approaches avoid causality violations by blocking execution, but often suffer from scalability limitations due to idle compute resources. Optimistic approaches allow speculative execution followed by rollback, but introduce memory overhead due to state snapshot storage.

Recent work has explored speculative synchronization methods to improve scalability. An-delfinger and Uhrmacher propose synchronous speculative simulation for tightly coupled agents, demonstrating improved performance in distributed simulations while highlighting the challenges of maintaining consistency across partitions [1].

Similarly, distributed simulation frameworks for digital twins rely on optimistic PDES engines to simulate large-scale systems, but still face synchronization overhead and state consistency challenges [2]. These findings reinforce the need for lightweight optimistic synchronization strategies.

In spatial simulations, distributed cellular automata remain a primary modeling approach for complex systems. Recent research on parallel cellular automata emphasizes boundary coordination and load balancing challenges in high-resolution spatial simulations [3]. Despite these advances,

existing approaches continue to rely on rollback mechanisms or heavy synchronization protocols, leaving room for deterministic, snapshot-free optimistic approaches.

3. Proposed Solution

The proposed solution implements a protocol that utilizes "Asynchronous Rule-Based Coupling." Instead of saving historical memory snapshots to facilitate rollbacks, the protocol assumes boundary integrity during the compute phase.

Post-compute, nodes exchange boundary state hashes. In the event of a conflict (e.g., divergent phase-change results at a shared (micro) scale), the system invokes a deterministic priority hierarchy. This hierarchy functions as a tie-breaker rule set (e.g., Solid > Liquid > Gas), allowing both nodes to converge on the same physical state independently. This approach transforms the distributed conflict from a network negotiation problem into a local deterministic resolution.

Feasibility and Implementation: The system will be implemented as a master-worker architecture. Communication will be handled via gRPC or UDP to test protocol resilience under varying network conditions. The implementation's novelty lies in its snapshot-free design, which is specifically optimized for nodes with restricted local memory. Evaluation will be conducted through empirical throughput testing, comparing this deterministic optimistic model against traditional conservative blocking.

4. Deliverables

1. **Project Report:** Technical analysis of boundary throughput(tick rate) and state convergence rates under varying network latency.
2. **Presentation Slides:** Research project presentation.
3. **Source Code:** Implementation of the worker and master nodes, including the conflict resolution engine.
4. **Technical Demonstration:** Demonstration(video) of independent state convergence across distributed partitions during high-frequency material interactions.

5. Timeline

- **Week 1:** Implementation of the spatial grid logic and the deterministic conflict resolution matrices.
- **Week 2:** Development of the gRPC-based worker orchestration and asynchronous hash-exchange protocol.
- **Week 3:** Integration of the optimistic coupling logic and implementation of network stress-testing tools.
- **Week 4:** Empirical data collection, throughput benchmarking, and final report compilation.

References

- [1] P. Andelfinger and A. Uhrmacher, "Synchronous speculative simulation of tightly coupled agents in continuous time on CPUs and GPUs" (<https://journals.sagepub.com/doi/10.1177/00375497231158930>)

- [2] Z. Huang et al., “Distributed Simulation for Digital Twins of Large-Scale Real-World DiffServ-Based Networks” arXiv preprint arXiv:2405.20815, 2024.
- [3] W. Xia et al., “Dynamic Load Balancing Based on Hypergraph Partitioning for Parallel Geospatial Cellular Automata Models” ISPRS International Journal of Geo-Information, 2025. (<https://www.mdpi.com/2220-9964/14/3/109>)
- [4] R. M. Fujimoto, “Parallel Discrete Event Simulation” ACM Computing Surveys, vol. 22, no. 1, pp. 1–45, 1990. <https://dl.acm.org/doi/10.1145/84537.84545>
- [5] C. D. Carothers, D. Bauer, and S. Pearce, “ROSS: A High-Performance, Low-Memory, Modular Time Warp System” Journal of Parallel and Distributed Computing, 2002. <https://doi.org/10.1006/jpdc.2001.1805>